

Data Structures

Lecture-01

What is a Data Structure?

A **Data Structure** is a method of organizing and storing data in a computer so that it can be accessed and modified efficiently.

Key Point:

Choosing the appropriate data structure improves the performance of an algorithm by making operations such as searching, insertion, deletion, and sorting faster.

Real-Life Analogy

Think of a **library**.

- If books are placed randomly, finding a book takes a long time.
- If books are arranged by subject or title, they can be found quickly.

Similarly,

Well Organized Data $\implies$ Fast Programs

Importance (Need) of Data Structures

Data structures help us to:

1. Store data efficiently.
2. Access and search data quickly.
3. Insert and delete data easily.
4. Reduce memory usage.
5. Improve the overall performance of programs.

Types of Data Structures

Data structures are broadly classified into two categories.

1. Linear Data Structures

In a **Linear Data Structure**, each element is connected in a sequence. Except for the first and last elements,

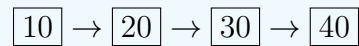
- Every element has exactly one predecessor.

- Every element has exactly one successor.

Examples

- Array
- Linked List
- Stack
- Queue

Visualization



2. Non-Linear Data Structures

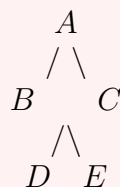
In a **Non-Linear Data Structure**, elements are not arranged in a single sequence. An element may have

- More than one predecessor, or
- More than one successor.

Examples

- Trees
- Graphs

Visualization



Linear vs Non-Linear Data Structures

Feature	Linear	Non-Linear
Organization	Sequential	Hierarchical/Network
Relationship	One-to-One	One-to-Many or Many-to-Many
Traversal	Simple	Comparatively Complex
Examples	Array, Stack, Queue, Linked List	Tree, Graph

Basic Operations on Data Structures

Operation	Meaning
Traversal	Visiting each element exactly once.
Insertion	Adding a new element.
Deletion	Removing an existing element.
Searching	Finding a specific element.
Sorting	Arranging elements in ascending or descending order.
Merging	Combining two similar data structures into one.

Lecture Summary

- A data structure organizes data efficiently.
- Choosing the correct data structure improves algorithm performance.
- Data structures are classified into:
 - Linear Data Structures
 - Non-Linear Data Structures
- Common operations include:
 - Traversal
 - Insertion
 - Deletion
 - Searching
 - Sorting
 - Merging

Remember

Good Data Structure $\implies$ Efficient Algorithm $\implies$ Fast Program